

Distributional Forecasting for Financial Time Series with Power-Law Behaviour

Fraser Buxton

April 20, 2026



Abstract

Financial log-returns, particularly those of cryptocurrencies, demonstrate substantial challenges for probabilistic forecasting through their high volatility and complexity. They are known, however, to exhibit meaningful empirical phenomena, such as volatility clustering and power-law tails. This study investigates the use of the GARCH(1,1) to capture these characteristics, with emphasis on the asymptotic tail behaviour.

Using Kesten's stochastic recurrence equation as a theoretical framework, the conditions for strict stationarity in the GARCH(1,1) process are established and it is rigorously proven that the resulting stationary process exhibits power-law tails. In this case, it is demonstrated that the tail index of the stationary process is dependent only on the parameters and the innovation distribution. An explicit expression for this value is derived under Gaussian innovations, as well as a practical method for computing it.

Empirically, this formulation is applied to a sample of 1000 daily Bitcoin log-returns. Multiple mean specifications, including ARMA(1,1), are assessed recursively using a rolling sample of 500 data points, both with and without GARCH-based volatility modelling. Each series of forecasts is evaluated using proper scoring criteria and is benchmarked against its mean-only counterpart by a Diebold-Mariano significance test. The results support the GARCH approach to consistently and significantly improve predictive performance across all mean specifications.

Moreover, this paper explores the impact of adding constraints to the tail index based on empirically estimated values. This method establishes a decrease in local accuracy but an increase in global calibration, introducing a trade-off between the two. The results demonstrate meaningful implications for econometric tail-risk applications and is of practical relevance when defining a model framework.

Contents

1	Empirical Phenomena for Financial Time Series	4
1.1	Stylised Facts	4
1.2	GARCH Processes	5
1.3	The Tails	7
2	Theoretical GARCH Properties	10
2.1	Stationarity	11
2.2	Power-Law Tails	14
2.3	Tail Indices	18
3	Distributional Forecasting using GARCH	22
3.1	Model Evaluation	24
3.2	Significance Testing	26
3.3	Tail Index Estimation	27
4	Discussion and Future Work	32

1 Empirical Phenomena for Financial Time Series

1.1 Stylised Facts

Modelling the prices of financial assets is fundamental for institutes to develop an understanding in the structure and the dependencies present in the market. By observing and predicting the evolution of the volatility, financial agents are able to quantify and manage risk, develop algorithmic trading strategies and derive arbitrage-free prices for financial derivatives. Together with market makers, this contributes to maintaining market accessibility and liquidity through continuous and fair pricing. If one were to believe in the 'efficient market hypothesis' [1], consistent outperformance of the market would be theoretically impossible, so managing and minimising risk becomes an important factor for most financial agents.

Fluctuations in asset prices are often due to unpredictable events (political and economic news, investor behaviour, for example). Since these factors can not be predicted deterministically, it is often useful to model asset prices as stochastic processes. Early empirical evidence was proposed by Fama [2], who documented the validity of the random-walk hypothesis. The Black-Scholes model for derivative pricing [3] also relies on the assumption that prices follow a geometric Brownian motion.

Many econometric methods require stationarity in the data (see [4]), but asset prices often exhibit significant positive drift. Thus, it is appropriate to consider log-returns to induce approximate stationarity in the mean. Denoting $(P_t)_{t \in \mathbb{Z}}$ as the price of a specific asset at time t , the corresponding log-returns process $(Y_t)_{t \in \mathbb{Z}}$,

$$Y_t = \ln(P_t) - \ln(P_{t-1}) = \ln\left(\frac{P_t}{P_{t-1}}\right), \quad \forall t \in \mathbb{N},$$

transforms the data to have an approximate mean of zero and an increased stability in the variance. Exemplarily, we refer to the log-returns of daily Bitcoin (BTC-USD) prices as a representation of the newer and lesser researched cryptocurrency market. Bitcoin was chosen due to its dominant market capitalisation and high liquidity.

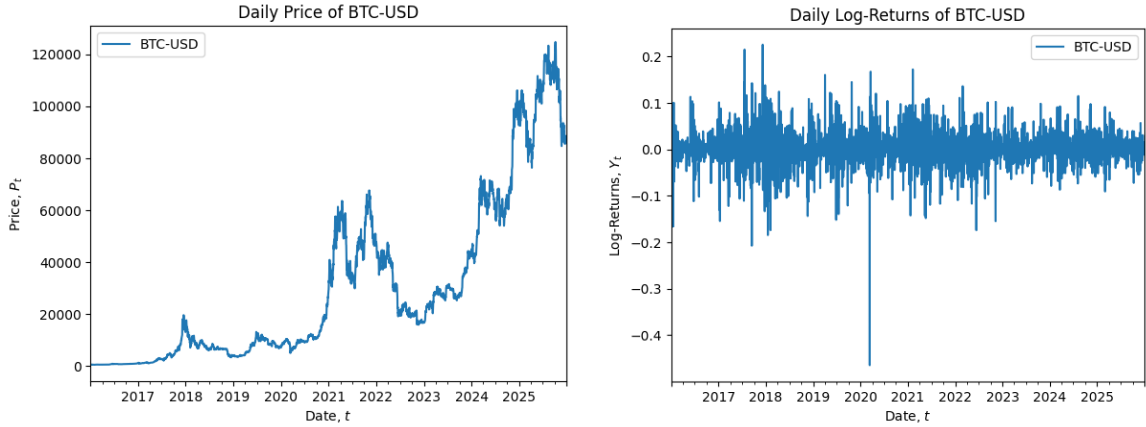


Figure 1: Left: Daily Bitcoin (BTC-USD) price P_t over the period from 1st January 2016 to 31st December 2025 accessed through Python's yfinance [5] package. Right: Daily Bitcoin (BTC-USD) log-returns Y_t over the same period.

Figure 1 emphasises the importance of using log-returns in predictive modelling. The price

time series shows significant positive drift and clear non-stationarity. The log-returns fluctuate approximately around zero and the volatility is much more manageable; while large spikes are still present, they occur more consistently. This stabilisation makes log-returns more suitable for econometric modelling.

The empirical properties of financial log-returns were explored in depth by Cont [6] and were presented as a number of 'stylised facts'. These facts have become principle in statistical analysis of financial data and are summarised as follows:

- **Absence of Autocorrelations:** Linear autocorrelations of the log-returns are often negligible.
- **Volatility Clustering:** High-volatility events (e.g. market crashes) tend to group at points in time.
- **Long-Term Memory in Volatility:** Correlations in the volatility often decay slowly over time.
- **Heavy Tails:** The unconditional distribution of the log-returns exhibit asymptotic power-law behaviour with finite variance.

Thus, replicating such phenomena within a predictive model is advisable and can enhance performance. The GARCH process is a popular example of this and will be an important tool for the probabilistic forecasting in this thesis.

1.2 GARCH Processes

In an attempt to introduce a stochastic process which expresses long-term dependence in the volatility, Engle [8] developed a model which calculated volatilities recurrently using residuals from the previous time steps. This was deemed the Autoregressive Conditional Heteroskedasticity (ARCH(p)) process,

$$X_t = \sigma_t Z_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i X_{t-i}^2, \quad \forall t \in \mathbb{Z},$$

and was extended further by Bollerslev [9], who introduced a term to consider the previous volatilities. The Generalised Autoregressive Conditional Heteroskedasticity (GARCH(p,q)) model,

$$X_t = \sigma_t Z_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i X_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2, \quad \forall t \in \mathbb{Z},$$

has numerous significant applications within econometrics, including risk management [10], derivative pricing [11] and volatility forecasting [9], the latter of which will be explored from theoretical and practical standpoint in this paper. By simulating data from an arbitrary GARCH(p,q) model for a comparison with the Bitcoin log-return sample, we aim to explore empirically the properties that the process exhibits and how this is representative of financial log-returns.

The presented stylised facts introduce the idea of autocorrelations,

$$\rho(\tau) = \mathbb{E}(X_t X_{t-\tau}),$$

where $(X_t)_{t \in \mathbb{Z}}$ is a stochastic process and $\tau \in \mathbb{N}_0$ a specific time lag. An appropriate empirical estimate [12] of this statistic is

$$\hat{\rho}(\tau) = \text{corr}(X_t, X_{t-\tau}),$$

where $\text{corr}()$ denotes the sample correlation. We develop empirical autocorrelation estimates for the simulated and true samples to assess their significance.

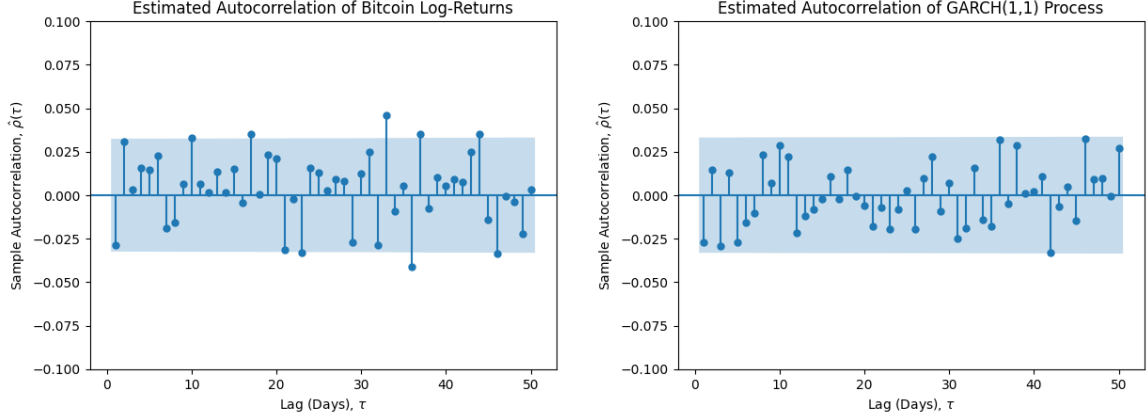


Figure 2: Empirical estimates of autocorrelation function for a lag range of 0 to 50 applied to the daily log-returns of Bitcoin over 10 years (Left) and simulated data from a GARCH(1,1) model with parameters $\alpha_0 = 0.0001, \alpha_1 = 0.1035, \beta_1 = 0.8650$ (Right).

Both plots in Figure 2 are overlaid with a 95% confidence region for the hypothesis:

$$H_0 : \hat{\rho}(\tau) = 0, \quad H_1 : \hat{\rho}(\tau) \neq 0$$

and show mostly insignificant values over the range lags. Some values, however, are significant enough to reject the null hypothesis that no autocorrelations exist and stimulates the existence of some small dependence in both the log-returns and the illustrative sample. This is due to underlying factors such as estimation noise or microstructural features in the cryptocurrency market.

The absence of autocorrelations would support the weak form efficient market hypothesis [1] through the lack of predictability in the returns. In principle, significant and persistent autocorrelations could be exploited to create trading strategies with risk-free profit. The plots suggest deviations in both the Bitcoin market and the simulated GARCH data consistent with mild inefficiencies in the sense of the hypothesis, though these are not necessarily economically exploitable when taking into account model uncertainty and transaction costs.

Absence of autocorrelations, however, does not imply independence as this would require no autocorrelations in any non-linear mapping of the log-returns. The persistence,

$$\rho_2(\tau) = \mathbb{E}(X_t^2 X_{t-\tau}^2),$$

for example, which can be estimated empirically,

$$\hat{\rho}_2(\tau) = \text{corr}(X_t^2, X_{t-\tau}^2),$$

exhibits significant positive values that provide evidence for some non-linear dependence. This quantitative measure is a great representation of volatility clustering as it is induced by memory in the magnitude of the historic data, not the sign.

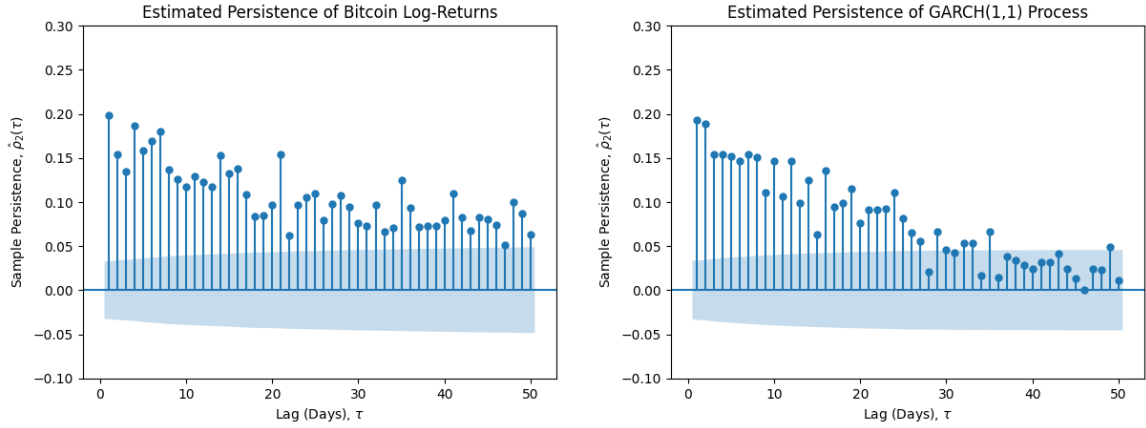


Figure 3: Empirical estimates of persistence for a lag range of 0 to 50 applied to the daily log-returns of Bitcoin over 10 years (Left) and simulated data from the squared GARCH(1,1) model with parameters $\alpha_0 = 0.0001, \alpha_1 = 0.1035, \beta_1 = 0.8650$ (Right).

The plots in Figure 3 contain 95% confidence intervals for the existence of non-zero persistence values and both show similar significant positive values that support this hypothesis. In turn, this provides evidence for the existence of volatility clustering. As the lag increases, the magnitude of this value slowly decreases. This slow decaying persistence, by definition, implies the existence of long-term dependence in the volatility. A paper by Cont, Potters and Bouchaud [13] showed that for the case of financial log-returns this decay can be well-represented by power-:

$$\rho_2(\tau) \sim \frac{A}{\tau^\beta},$$

for some constants $A, \beta \in \mathbb{R}$. This is a known property of the ARCH/GARCH processes, often deemed the 'ARCH effect' in econometric literature [8].

1.3 The Tails

The unconditional distribution of financial log-returns tend to show asymptotic tails heavier than the Gaussian distribution. It follows contextually that extreme events (market crashes etc.) happen more often than one would expect with a normally distributed model.

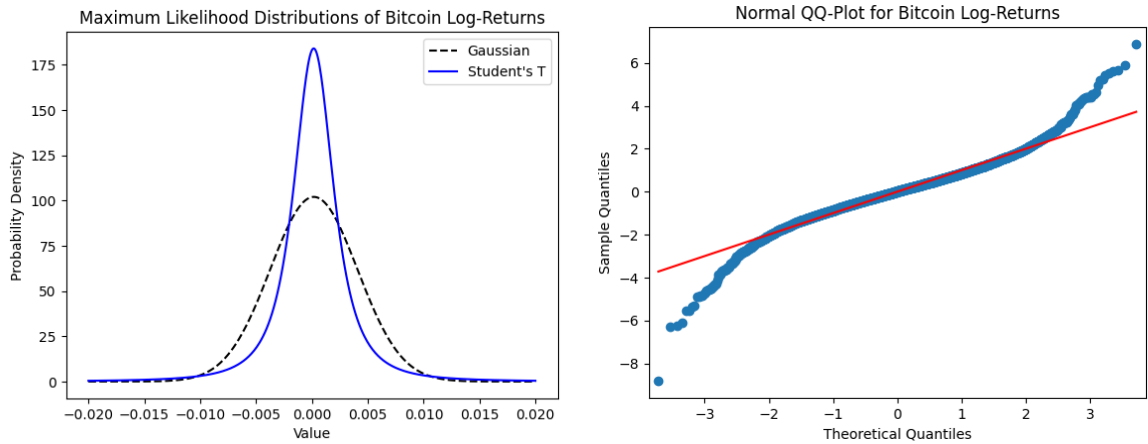


Figure 4: Left: Bitcoin daily log-returns fitted with maximum likelihood estimates of both Gaussian and Student's-t models. Right: Normal QQ-plot of Bitcoin daily log-returns

Both plots in Figure 4 support the speculation of asymptotic heaviness in the log-returns. The Student's-t distribution has much greater flexibility than the Gaussian over the heaviness of its tails via the degree of freedom parameter, ν . Maximum likelihood estimation produced a degree of freedom of $\hat{\nu} = 2.049$ (much smaller than that of the Gaussian distribution, $\nu = \infty$). This can be distinguished by the distribution curve of the Student's t having a much larger peak and bigger tails as we approach $\pm\infty$. The normal Q-Q plot show much smaller values for the negative theoretical quantiles that one would expect, and similarly much larger values for the positive, both giving the implication of heavy asymptotic behaviour.

This behaviour in financial log-returns was an idea introduced by Mandelbrot [14], who conjectured that sample moments become either divergent or erratic for large n when the true theoretical moment is infinite, and converges otherwise. Thus came the notion of the tail index κ , a quantitative measure for the heaviness of the tails and the largest positive number such that the κ -th moment is finite: the greater the tail index, the thinner the tail. For instance, the tail index of a Student's-t process is equal its degree of freedom, $\kappa = \nu$ [15].

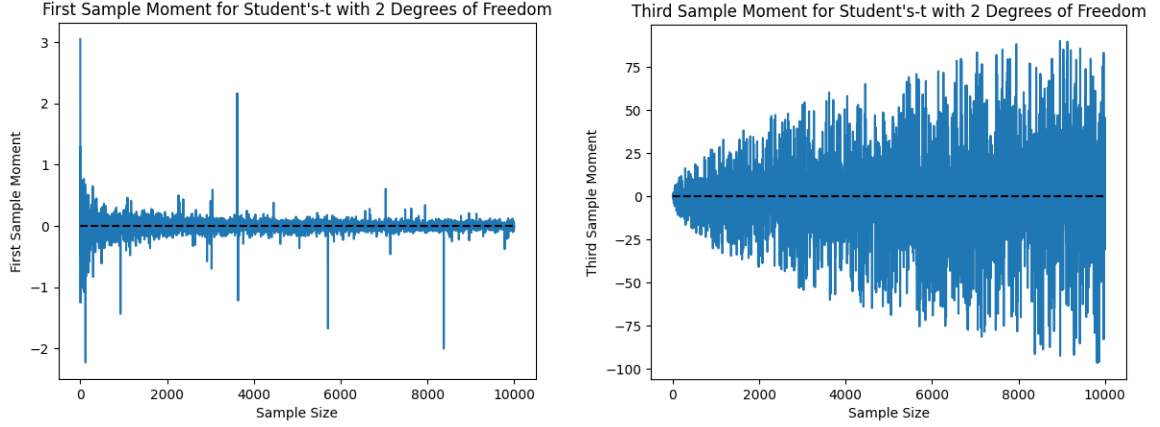


Figure 5: Empirical estimates of the first (Left) and third (Right) moments for simulated standard Student's-t distributed data with 2 degrees of freedom

The sample process used in Figure 5 is a standard Student's-t distribution with degree of freedom $\nu = 2$. This process has a known tail index of 2, equivalent to the number of degrees of freedom. Thus, we would expect divergent sample moments for any greater than (and including) the second. The first moment appears to show a convergence to 0 as n increases, agreeing with the known true value of 0. The third moment appears erratic and divergent, which admits to Mandelbrot's conjecture as the true value is known to be infinite by the tail index.

It is in fact the case that the unconditional tails of financial log-returns exhibit power-law behaviour [14],

$$\mathbb{P}(Y_t > y) \sim c_+ y^{-\kappa}, \quad \mathbb{P}(Y_t < -y) \sim c_- y^{-\kappa}, \quad y \rightarrow \infty,$$

where $c_{\pm}$ are constants and κ is the tail index of the distribution. This is a stronger property than just 'heavy tails' and it will be proven rigorously in Chapter 2 that it is a property also held by the GARCH process.

The similar properties of the two distributions motivates the appropriateness of using GARCH as a predictive model for the log-returns. Accordingly, this thesis will be structured as followed:

- Chapter 2 will analyse the inherent stylised facts within the GARCH model, verifying theoretically the conditions for the stationarity of the process and the existence of the power-law behaviour in the tails. This will be done by embedding the model in the framework of the 'Stochastic Recurrence Equation' [16],

$$X_t = A_t X_{t-1} + B_t, \quad t \in \mathbb{Z},$$

which is interesting in its own right, with applications in telecommunication and fractal images [7].

- Chapter 3 will apply the GARCH model as a volatility forecaster for the predictions of the next day log-returns of Bitcoin. A range of models, with and without GARCH, will be used to make such probabilistic forecasts and evaluated using proper scoring methods [17]. This invites the primary area of investigation of this research paper, whether (1) GARCH implementation improves predictability in Bitcoin returns forecasts and (2) the effects of constraining the tail-index of this distribution.

2 Theoretical GARCH Properties

The auto-regressive conditional heteroskedasticity (ARCH) model was presented by Engle [8] in an attempt to simulate a long-term memory through persistent volatility clustering. This is done with a separate volatility process which includes a term for the residuals of the previous time steps.

Definition 2.1 (ARCH(p) Process [8]). *Let $(Z_t)_{t \in \mathbb{Z}}$ denote an i.i.d. sequence of symmetrical, discrete-time random variables with mean zero and variance one. A random process $(X_t)_{t \in \mathbb{Z}}$ is called an ARCH(p) process if*

$$X_t = \sigma_t Z_t,$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i X_{t-i}^2,$$

where α_i are non-negative constants, more specifically

$$p \in \mathbb{N}, \quad \alpha_0, \alpha_i \in \mathbb{R}$$

$$\alpha_0 > 0, \quad \alpha_i \geq 0, \quad i = 1, \dots, p$$

The sum in the volatility process allows the conditional variance to change with t based on the errors of the previous time steps. This, however, does not take into account the previous volatilities in the sequence. Therefore, adding a lag term generalises the model further and was done by Bollerslev [9] with the GARCH model.

Definition 2.2 (GARCH(p, q) Process [9]). *Let $(Z_t)_{t \in \mathbb{Z}}$ denote an i.i.d. sequence of symmetrical, discrete-time random variables with mean zero and variance one. A random process $(X_t)_{t \in \mathbb{Z}}$ is called a GARCH(p, q) process if*

$$X_t = \sigma_t Z_t,$$

$$\sigma_t^2 = \alpha_0 + \sum_{i=1}^p \alpha_i X_{t-i}^2 + \sum_{j=1}^q \beta_j \sigma_{t-j}^2,$$

where α_i, β_i are non-negative constants, more specifically

$$p, q \in \mathbb{N}, \quad \alpha_0, \alpha_i, \beta_j \in \mathbb{R},$$

$$\alpha_0 > 0, \quad \alpha_i, \beta_j \geq 0, \quad i = 1, \dots, p, \quad j = 1, \dots, q,$$

For the purpose of this thesis, the research will be restricted only to the GARCH(1,1) process. The properties of this process are inherited directly from the Stochastic Recurrence Equation (SRE) [16].

Definition 2.3 (Stochastic Recurrence Equation [16]). *Let $((A_t, B_t))_{t \in \mathbb{Z}}$ denote an i.i.d. sequence with A_t as $d \times d$ matrices with real-valued entries and B_t as $\mathbb{R}^d$ -valued vectors for $d \in \mathbb{N}$. Then the process $(X_t)_{t \in \mathbb{Z}}$ takes the form of the stochastic recurrence equation if*

$$X_t = A_t X_{t-1} + B_t.$$

The volatility process of the GARCH(1,1) model can be embedded into this framework,

$$\sigma_t^2 = \alpha_0 + \alpha_1 Z_t^2 \sigma_{t-1}^2 + \beta_1 \sigma_{t-1}^2$$

$$= \underbrace{\alpha_0}_{B_t} + \underbrace{(\alpha_1 Z_t^2 + \beta_1)}_{A_t} \sigma_{t-1}^2,$$

which tells us that the properties of the stochastic recurrence equation succeed to the volatility process (σ_t^2) of the GARCH(1,1) process. Consequently, the existence of a stationary solution and its properties follow from general results from the equation.

2.1 Stationarity

A process is strictly stationary if the joint probability distribution, and therefore its statistical properties, are identical over shifts in time. This is a useful tool in time series analysis as it ensures that the past and future behaviour of the series have equivalent distributions and that the historic data contains useful information for predictive forecasts. As stated, taking log-returns of asset prices induces approximate stationarity by stabilising the mean and variance.

To support inference on the stationarity of the stochastic recurrence equation, Goldie and Maller [18] considered the outer iteration of the sequence:

$$\begin{aligned} X_t &= A_t X_{t-1} + B_t \\ &= A_t A_{t-1} X_{t-2} + (B_t + A_t B_{t-1}) \\ &\vdots \\ &= \Pi_{1,t} X_0 + \sum_{i=1}^t \Pi_{i+1,t} B_i, \end{aligned}$$

$\forall t \in \mathbb{N}_0$, where the notation

$$\Pi_{i,j} = \begin{cases} A_i \dots A_j & \text{if } i \leq j \\ 1 & \text{if } i > j \end{cases}$$

$\forall i, j \in \mathbb{N}_0$ is used for convenience.

From this representation, X_T depends only the sequence $(X_t)_{t \in \mathbb{Z}}$ through the initial value X_0 . This shows that the process defines a time-homogenous Markov chain. Therefore, the stationarity is conditioned on the existence of the stationary distribution of such. As presented by Buraczewski, Damek and Mikosch [7], the SRE is very closely related to the identity in distribution

$$X \stackrel{d}{=} AX + B,$$

where $(A, B) \in \mathbb{R}^{d \times d} \times \mathbb{R}^d$ is independent of the $\mathbb{R}^d$ -valued X . In the case where the identity is satisfied for some generic elements (A_0, B_0) and X_0 , the distribution $\mathbb{P}_{X_0}$ of X_0 is the stationary distribution of this Markov chain. Hence, one can treat the SRE as a fixed point equation, with the process $(X_t)_{t \in \mathbb{Z}}$ exhibiting stationarity if and only if the equation permits a strictly stationary solution. This is useful for analysing convergence and stationarity.

These conditions for the existence of this solution can be found by considering the inner iteration, that is a sequence of random affine maps $(X_t(\cdot))_{t \in \mathbb{N}}$ with X_0 a random variable independent of the series,

$$\begin{aligned} X_t(X_0) &= X_{t-1}(A_t X_0 + B_t) \\ &= X_{t-2}(A_{t-1}(A_t X_0 + B_t) + B_{t-1}) \\ &= X_{t-2}(A_{t-1} A_t X_0 + A_{t-1} B_t + B_{t-1}) \\ &\vdots \\ &= \Pi_{1,t} X_0 + \sum_{i=1}^t \Pi_{1,i-1} B_i, \end{aligned}$$

and thus the existence of a stationary solution relies on the almost-sure convergence of the infinite series above as $t \rightarrow \infty$. Equivalent notions of this infinite series were developed by Goldie and Maller [18] in their paper

Proposition 2.4 (Convergence [18]). *Consider an i.i.d. $\mathbb{R}^2$ -valued sequence $((A_t, B_t))_{t \in \mathbb{Z}}$ with*

- $\mathbb{P}(B_t = 0) < 1$
- $\mathbb{P}(A_t = 0) = 0$

Then the following are equivalent

- $\sum_{i=1}^t \Pi_{1,i-1} B_i$ converges absolutely almost surely
- $\Pi_{1,t-1} B_t \xrightarrow{a.s.} 0$ as $t \rightarrow \infty$

Both equivalent cases give the implication of

$$\Pi_{t-1} B_t + \sum_{i=1}^t \Pi_{1,i-1} B_i \xrightarrow{a.s.} \sum_{i=1}^t \Pi_{1,i-1} B_i < \infty,$$

and thus are equivalent to the almost sure convergence of the infinite series representation. Therefore, this supports the existence of a strictly stationary solution to the stochastic recurrence equation. Using this knowledge, it is possible to derive specific sufficient and necessary conditions for existence of this stationary solution.

Theorem 2.5 (Stationarity [7]). *a) Consider an i.i.d. $\mathbb{R}^2$ -valued sequence $((A_t, B_t))_{t \in \mathbb{Z}}$ and assume that one of the following conditions holds*

1. $\mathbb{P}(A_t = 0) > 0$;
2. $\mathbb{P}(A_t = 0) = 0, -\infty \leq \mathbb{E}(\log |A_t|) < 0, \mathbb{E}(\log^+ |B_t|) < \infty$.

then there exists an almost-sure unique causal ergodic strictly stationary solution to the stochastic recurrence equation.

b) Moreover, no strictly stationary causal solution exists if either of the conditions

1. $\mathbb{P}(A_t = 0) = 0, \mathbb{P}(B_t = 0) < 1, 0 \leq \mathbb{E}(\log |A_t|) < \infty$
2. $\mathbb{E}(\log |A_t|) > -\infty, \mathbb{E}(\log^+ |B_t|) = \infty$.

hold.

Therefore, for the case of the GARCH(1,1) framework, that is,

$$A_t = \alpha_1 Z_t^2 + \beta_1, \quad B_t = \alpha_0,$$

we must derive the combinations of $\alpha_0, \alpha_1, \beta_1$ and Z_t such that the conditions in Theorem 2.5 hold. Here, $A_t \geq 0$ almost surely and $B_t > 0$ is due to the restrictions on the parameters in the GARCH definition. That is, since $\alpha_1, \beta_1 \geq 0$,

$$\begin{aligned} \mathbb{P}(Z_t^2 \geq 0) &= 1 \Rightarrow \mathbb{P}(\alpha_1 Z_t^2 \geq 0) = 1 \\ &\Rightarrow \mathbb{P}(A_t = \alpha_1 Z_t^2 + \beta_1 \geq 0) = 1, \end{aligned}$$

and

$$B_t = \alpha_0 > 0.$$

Theorem 2.6 (Stationarity of GARCH(1,1)). *Let $(X_t)_{t \in \mathbb{Z}}$ be GARCH(1,1) process with parameters $\alpha_0, \alpha_1, \beta_1 > 0$. An almost-surely unique causal ergodic strictly stationary solution exists if and only if*

$$\mathbb{E}(\log(\alpha_1 Z_t^2 + \beta_1)) < 0$$

Proof. It suffices to prove that the volatility process permits to the conditions in Theorem 2.5 as the existence of the stationary solution for X_t follows. First note that $\alpha_1, \beta_1 > 0$ and $Z_t^2 \geq 0$, which gives

$$A_t = \alpha_1 Z_t^2 + \beta_1 \geq \beta_1 > 0 \Rightarrow \mathbb{P}(A_t = 0) = 0.$$

We now verify the remaining conditions. For sufficiency, it is required that

$$-\infty \leq \mathbb{E}(\log |A_t|) < 0, \quad \mathbb{E}(\log^+ |B_t|) < \infty.$$

Since $B_t = \alpha_0 > 0$ is strictly positive and constant, the second inequality is met automatically by

$$\mathbb{E}(\log^+ |B_t|) = \mathbb{E}(\log^+ \alpha_0) < \infty.$$

Then, if it is assumed that

$$\mathbb{E}(\log(\alpha_1 Z_t^2 + \beta_1)) = \mathbb{E}(\log A_t) < 0,$$

then the almost sure non-negativity of A_t gives

$$A_t = |A_t| \implies \mathbb{E}(\log(\alpha_1 Z_{t-1}^2 + \beta_1)) = \mathbb{E}(\log A_t) = \mathbb{E}(\log |A_t|) < 0,$$

and therefore the second sufficiency condition in Theorem 2.5a has been met and the unique stationary solution exists.

If, on the other hand, we assume the negation of the condition,

$$\mathbb{E}(\log(\alpha_1 Z_t^2 + \beta_1)) = \mathbb{E}(\log A_t) = \mathbb{E}(\log |A_t|) \geq 0,$$

then since $A_t = \alpha_1 Z_t^2 + \beta_1$ and Z_t has finite moments, it follows that $\log A_t$ is integrable and $\mathbb{E}(\log A_t) < \infty$. Then since we know that

$$B_t = \alpha_0 > 0 \implies \mathbb{P}(B_t = 0) = \mathbb{P}(\alpha_0 = 0) = 0,$$

then our condition is also necessary by the first condition in Theorem 2.5b. $\square$

Theorem 2.6 implies that the stationarity of the process depends only on the parameters α_1, β_1 and the distribution of the innovation variable Z_t . Though this method is precise, it may be computationally complex due to the integral within the expectation. However, it is possible to find a sufficient condition for the solution which does not depend on the innovation distribution. The concavity of the log function allows Jensen's inequality to be applied,

$$\begin{aligned} \mathbb{E}(\log(\alpha_1 Z_t^2 + \beta_1)) &\leq \log(\mathbb{E}(\alpha_1 Z_t^2 + \beta_1)) \\ &= \log(\alpha_1 \mathbb{E}(Z_t^2) + \beta_1) \\ &= \log(\alpha_1 + \beta_1), \end{aligned}$$

from the assumption that $\mathbb{E}(Z_t^2) = 1$. This motivates the sufficiency condition:

$$0 < \alpha_1 + \beta_1 < 1 \Rightarrow \mathbb{E}(\log(\alpha_1 Z_t^2 + \beta_1)) \leq \log(\alpha_1 + \beta_1) < 0.$$

and thus a stationary solution exists no matter the innovation. It will be shown in the next section that this condition ensures covariate stationarity (finite second moment). It has become general convention in the econometrics community to assume this condition to ensure stationarity, but this condition is not necessary so may exclude some stationary processes.

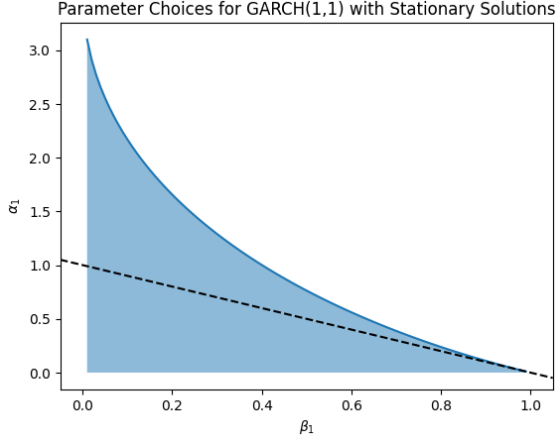


Figure 6: Curve of the function $\mathbb{E}(\log(\alpha_1 Z_t^2 + \beta_1)) = 0$ with Gaussian innovations over various values of α_1 and β_1 . The shaded area beneath the curve represents the values such that a stationary solution to the GARCH(1,1) process exists. The dashed line represents the equation $\alpha_1 + \beta_1 = 1$. Notice that any parameter pairs representing $\alpha_1 + \beta_1 < 1$ lie within the shaded region.

2.2 Power-Law Tails

The asymptotic tails of the conceptual distribution of Y_t of the financial log-returns were shown by Mandelbrot [14] to exhibit power-law behaviour,

$$\mathbb{P}(Y_t > y) \sim c_+ y^{-\kappa}, \quad \mathbb{P}(Y_t < -y) \sim c_- y^{-\kappa}, \quad y \rightarrow \infty,$$

for some constants $c_{\pm}$ and tail index κ . Therefore, it is sensible to desire models that act similarly. It is possible to prove that the GARCH configuration displays adequate similarities through the use of the stochastic recurrence equation. In his paper, Kesten [16] presented that under mild assumptions the unique solution to the equation, if it exists, expresses power-law properties. Subsequently, this was built on by Goldie [19] who derived the values for the constants.

Theorem 2.7 (Kesten-Goldie Theorem [16] [19]). *Assume that the following conditions hold:*

- $A \geq 0$ almost surely
- The conditional distribution of $\log A$ given $\{A \neq 0\}$ is non-arithmetic
- There exists $\alpha > 0$ such that
 - $\mathbb{E}(A^\alpha) = 1$
 - $\mathbb{E}(|B|^\alpha) < \infty$
 - $\mathbb{E}(A^\alpha \log^+ A) < \infty$
- $\mathbb{P}(Ax + B = x) < 1$ for all $x \in \mathbb{R}$

where $\log^+(x) = \max\{0, \log(x)\}$. Then the equation $X \stackrel{d}{=} AX + B$ has a solution X which is independent of (A, B) and there exists constants c_+, c_- such that $c_+ + c_- > 0$ and

$$\mathbb{P}(X > x) \sim c_+ x^{-\alpha}, \quad \mathbb{P}(X < -x) \sim c_- x^{-\alpha}, \quad x \rightarrow \infty$$

where

$$c_+ = \frac{1}{2\alpha m_\alpha} \mathbb{E}((AX + B)_+^\alpha - (AX)_+^\alpha)$$

$$c_- = \frac{1}{2\alpha m_\alpha} \mathbb{E}((AX + B)_-^\alpha - (AX)_-^\alpha)$$

and

$$m_\alpha = \mathbb{E}(A^\alpha \log A) > 0$$

This is the fundamental theorem in proving the existence of power-law tails in the case of GARCH(1,1). Recall that this context is applied using the stochastic recurrence equation,

$$\sigma_t^2 = A_t \sigma_{t-1}^2 + B_t, \quad A_t = \alpha_1 Z_t^2 + \beta_1, \quad B_t = \alpha_0,$$

as this corresponds to the volatility process of the GARCH foundations. Since the stationary solution coincides with the solution of the fixed point equation [7],

$$\sigma^2 \stackrel{d}{=} A\sigma^2 + B,$$

verifying the Kesten-Goldie conditions for (A, B) implies the existence of power law tails in the stationary process σ_t^2 . We first establish an upper bound for the function $\log^+(x) = \max\{0, \log(x)\}$.

Lemma 2.8. *For all $x, \varepsilon > 0$, the strict inequality*

$$\log^+(x) < \frac{x^\varepsilon}{\varepsilon}$$

holds.

Proof. Consider the function, for some $\varepsilon > 0$,

$$f(x) = \frac{x^\varepsilon}{\varepsilon} - \log x$$

and differentiate

$$f'(x) = x^{\varepsilon-1} - \frac{1}{x} = \frac{x^\varepsilon - 1}{x}.$$

Thus, the function is non-decreasing on the interval $[1, \infty)$ since $x^\varepsilon \geq 1$. Consider the boundary of this interval,

$$f(1) = \frac{1}{\varepsilon} - 0 = \frac{1}{\varepsilon} > 0.$$

Hence, for any $x \geq 1$,

$$f(x) \geq f(1) > 0 \Rightarrow \frac{x^\varepsilon}{\varepsilon} - \log x > 0$$

$$\Rightarrow \frac{x^\varepsilon}{\varepsilon} > \log x,$$

and therefore, by definition,

$$\log^+(x) = \begin{cases} \log(x) & \text{if } x > 1 \\ 0 & \text{if } x \leq 1, \end{cases}$$

which gives us

$$\log^+(x) < \frac{x^\varepsilon}{\varepsilon}.$$

□

Then, it can be shown that the conditions of Theorem 2.7 are satisfied when considering the GARCH(1,1) process, suggesting that the existence of a stationary solution implies that it embodies asymptotic power-law behaviour.

Theorem 2.9 (Power-Law Tails of GARCH(1,1)). *Consider the GARCH(1,1) model in the form of the stochastic recurrence equation:*

$$\sigma_t^2 = A_t \sigma_{t-1}^2 + B_t, \quad A_t = \alpha_1 Z_t^2 + \beta_1, \quad B_t = \alpha_0$$

Then the solution σ^2 , if it exists, is independent of (A, B) , where (A, B) is a generic independent copy of (A_t, B_t) in the fixed point equation $\sigma^2 \stackrel{d}{=} A\sigma^2 + B$ and satisfies the Kesten-Goldie conditions. Therefore, there exists a constant c_+ such that:

$$\mathbb{P}(\sigma^2 > x) \sim c_+ x^{-\alpha}, \quad x \rightarrow \infty$$

Proof. Let Z be a generic copy of the innovation sequence $(Z_t)_{t \in \mathbb{Z}}$, and define

$$A = \alpha_1 Z^2 + \beta_1, \quad B = \alpha_0.$$

Then (A, B) is independent of σ^2 . It suffices to verify the conditions of the Kesten–Goldie theorem for the identity in distribution $\sigma^2 = A\sigma^2 + B$.

Since $\alpha_1, \beta_1 \geq 0$, we have $A \geq 0$ almost surely. Furthermore, since Z has a continuous distribution and $A = \alpha_1 Z^2 + \beta_1$ is a continuous transformation, $\log A$ (under $A \neq 0$) has a non-arithmetic distribution as continuous distributions cannot be supported on a lattice.

To prove the existence of $\alpha > 0$ which satisfies the $\mathbb{E}(A^\alpha) = 1$ it suffices to make use of the Intermediate Value Theorem. Note that the assumed existence of a stationary solution gives

$$\mathbb{E}(\log A) < 0,$$

by Theorem 2.6. To begin, define the generating function for $p \geq 0$,

$$\begin{aligned} g(p) &= \mathbb{E}(A^p) \\ &= \mathbb{E}(e^{p \log A}), \end{aligned}$$

then consider the first order Taylor expansion around $p = 0$,

$$e^{p \log A} = 1 + p \log A + o(p^2),$$

and take expectations,

$$\mathbb{E}(e^{p \log A}) = 1 + p \mathbb{E}(\log(A)) + o(p^2).$$

Since by assumption $\mathbb{E}(\log(A)) < 0$, we can choose $p = \gamma$ small enough that

$$g(\gamma) = \mathbb{E}(A^\gamma) = 1 + \gamma \mathbb{E}(\log(A)) + o(\gamma^2) < 1.$$

Conversely consider, for any $c > 1$,

$$\begin{aligned} \mathbb{E}(A^p) &\geq \mathbb{E}(A^p \mathbb{I}_{\{A > c\}}) \\ &\geq c^p \mathbb{P}(A > c). \end{aligned}$$

Then as $p \rightarrow \infty$

$$\mathbb{E}(A^p) \geq c^p \mathbb{P}(A > c) \rightarrow \infty,$$

that is, $g(p) \rightarrow \infty$ as $p \rightarrow \infty$. Hence, since $g(p) : p \mapsto \mathbb{E}(A^p) = \mathbb{E}((\alpha_1 Z^2 + \beta_1)^p)$ is continuous, by the Intermediate Value Theorem there exists an $\alpha > \gamma > 0$ such that $\mathbb{E}(A^\alpha) = 1$. To prove the uniqueness of this α , see that

$$\begin{aligned}\mathbb{E}(A^p) &= \mathbb{E}(e^{p \log A}) \\ \frac{d}{dp} \mathbb{E}(A^p) &= \mathbb{E}(e^{p \log A} \log A) = \mathbb{E}(A^p \log A) \\ \frac{d^2}{dp^2} \mathbb{E}(A^p) &= \mathbb{E}(e^{p \log A} (\log A)^2) = \mathbb{E}(A^p (\log A)^2) > 0\end{aligned}$$

is a positive second derivative under assumptions such that differentiation and integration are interchangeable. Therefore, we have strict convexity on $\mathbb{R}$, so the solution is unique. Then, for such α it can be seen that

$$\mathbb{E}(|B|^\alpha) = \alpha_0^\alpha < \infty$$

and through use of Lemma 2.8 for some $\varepsilon > 0$

$$\begin{aligned}\log^+ A &\leq \frac{1}{\varepsilon} A^\varepsilon \Rightarrow A^\alpha \log^+ A \leq \frac{1}{\varepsilon} A^{\alpha+\varepsilon} \\ &\Rightarrow \mathbb{E}(A^\alpha \log^+ A) \leq \frac{1}{\varepsilon} \mathbb{E}(A^{\alpha+\varepsilon}) < \infty,\end{aligned}$$

by light tails and therefore finite moments of Z . Both conditions are required for the theorem.

Finally, note that when $x = 0$,

$$\mathbb{P}(Ax + B = x) = \mathbb{P}(B = 0) = \mathbb{P}(\alpha_0 = 0) = 0 < 1,$$

then for $x \neq 0$,

$$\mathbb{P}(Ax + B = x) = \mathbb{P}\left(A = \frac{x - \alpha_0}{x}\right) = 0,$$

by the continuity of A once again.

All the conditions for the Kesten-Goldie theorem are satisfied, so for the unique α that satisfies $\mathbb{E}(A^\alpha) = 1$,

$$\mathbb{P}(\sigma^2 > x) \sim c_+ x^{-\alpha}, \quad \mathbb{P}(\sigma^2 < -x) \sim c_- x^{-\alpha}, \quad x \rightarrow \infty,$$

with

$$c_- = \frac{1}{2\alpha m_\alpha} \mathbb{E}((A\sigma^2 + B)_-^\alpha - (A\sigma^2)_-^\alpha) = 0$$

by the almost sure non-negativity A, σ^2 and B . This reflects the fact that σ^2 is non-negative and so the left tail is trivial. $\square$

This theorem concludes that whenever a strictly stationary solution exists, the volatility process exhibits power-law tails. In particular, there exists an $\alpha > 0$ such that $\mathbb{E}(A^\alpha) = 1$ and

$$\mathbb{P}(\sigma^2 > x) \sim c_+ x^{-\alpha} \Rightarrow \mathbb{P}(\sigma > x) = \mathbb{P}(\sigma^2 > x^2) \sim c_+ x^{-2\alpha},$$

so σ has tail index $\kappa = 2\alpha$. Now, by the GARCH(1,1) definition $X = \sigma Z$ and Z is light-tailed. Thus, the tail index is preserved through multiplication of Z by standard results on products of random variables (Breiman's Lemma [20]),

$$\mathbb{P}(|\sigma Z| > x) \sim \mathbb{E}(|Z|^\kappa) \mathbb{P}(\sigma > x),$$

by the finiteness of moments of Z . Therefore, under the existence of a strictly stationary solution, the GARCH(1,1) process exhibits power-law tails with index $\kappa = 2\alpha$, where α is the unique solution to the equation $\mathbb{E}(A^\alpha) = 1$.

This result demonstrates that the GARCH(1,1) model generates heavy-tailed return distributions even when the innovations Z_t themselves are light-tailed. This behaviour arises from the stochastic recurrence equation, where the random coefficient $A_t = \alpha_1 Z_t^2 + \beta_1$ sometimes take large values, leading to persistent periods of high volatility. Since the solution is still inherently stationary, the process reflects a balance between mean reversion and occasional random spikes. This reproduces the empirically observed heavy-tailed behaviour of the financial log-returns, such as those observed in our Bitcoin sample.

2.3 Tail Indices

It was proven in the previous section that the tail index κ and thus the heaviness of the asymptotic tails of the GARCH(1,1) process depend on the choice of α_1, β_1 through the equation

$$\mathbb{E}(A_t^{\frac{\kappa}{2}}) = \mathbb{E}((\alpha_1 Z_t^2 + \beta_1)^{\frac{\kappa}{2}}) = 1$$

and the distribution of i.i.d. random variables $(Z_t)_{t \in \mathbb{Z}}$. The aim of this section is to manipulate this equality into a function of α_1, β_1 such that for a given innovation distribution (for the purposes of this paper Gaussian innovations will be used) the tail index can be computed directly. Foundationally, it is helpful to derive the distribution of $(\alpha_1 Z_t^2 + \beta_1)^{\frac{\kappa}{2}}$ with respect to our known distribution of Z_t . Consequently, this will allow the expectation of such to be computed accordingly. Firstly, a lemma is introduced to define the behaviour of the distribution when some arbitrary symmetrical random variable Z is squared.

Lemma 2.10. *Let Z be a continuous random variable with symmetrical distribution F_Z , that is $\forall t \in \mathbb{R}$*

$$F_Z(-t) = 1 - F_Z(t),$$

then the distribution of Z^2 can be defined by

$$F_{Z^2}(t) = \begin{cases} 2F_Z(\sqrt{t}) - 1 & \text{if } t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Proof. Note that, due to the non-negativity of a square, we have

$$F_{Z^2}(t) = \mathbb{P}(Z^2 \leq t) = 0 \quad \forall t < 0$$

For the case of $t \geq 0$ the distribution can be seen by

$$\begin{aligned} F_{Z^2}(t) &= \mathbb{P}(Z^2 \leq t) \\ &= \mathbb{P}(-\sqrt{t} \leq Z \leq \sqrt{t}) \\ &= \mathbb{P}(Z \leq \sqrt{t}) - \mathbb{P}(Z \leq -\sqrt{t}) \\ &= F_Z(\sqrt{t}) - F_Z(-\sqrt{t}) \\ &= 2F_Z(\sqrt{t}) - 1 \end{aligned}$$

where the final step was due to the symmetry of Z . This gives the distribution

$$F_{Z^2}(t) = \begin{cases} 2F_Z(\sqrt{t}) - 1 & \text{if } t \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

□

An extension of this lemma can be applied to determine the effect of the mapping $Y = (\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}}$ on the distribution of the arbitrary symmetrical stochastic process Z .

Proposition 2.11. *Let Z be a continuous random variable with symmetrical distribution F_Z . Then for $\alpha_1, \beta_1, \kappa > 0$ the distribution F_Y of $Y = (\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}}$ can be defined by*

$$F_Y(t) = \begin{cases} 2F_Z\left(\sqrt{\frac{t^{\frac{2}{\kappa}} - \beta_1}{\alpha_1}}\right) - 1 & \text{if } t \geq \beta_1^{\frac{\kappa}{2}} \\ 0 & \text{otherwise} \end{cases}$$

Proof. Firstly, we emphasise that $\alpha_1 Z^2 + \beta_1$ is almost surely non-negative

$$\begin{aligned} \mathbb{P}(Z^2 \geq 0) &= 1 \Rightarrow \mathbb{P}(\alpha_1 Z^2 \geq 0) = 1 \\ &\Rightarrow \mathbb{P}(\alpha_1 Z^2 + \beta_1 \geq 0) = 1 \end{aligned}$$

via the inherent non-negativity of a square and the constraints on α_1 and β_1 such that they are also non-negative. Consequently, it is known that the $\frac{\kappa}{2}$ th root is well defined so

$$\begin{aligned} F_Y(t) &= \mathbb{P}((\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}} \leq t) \\ &= \mathbb{P}(\alpha_1 Z^2 + \beta_1 \leq t^{\frac{2}{\kappa}}) \\ &= \mathbb{P}(Z^2 \leq \frac{t^{\frac{2}{\kappa}} - \beta_1}{\alpha_1}) \\ &= F_{Z^2}\left(\frac{t^{\frac{2}{\kappa}} - \beta_1}{\alpha_1}\right) \end{aligned}$$

where the division by α_1 leaves the inequality unchanged due to the non-negativity restriction. Hence, by Lemma 2.10 the distribution with respect to Z is

$$F_Y(t) = \begin{cases} 2F_Z\left(\sqrt{\frac{t^{\frac{2}{\kappa}} - \beta_1}{\alpha_1}}\right) - 1 & \text{if } \frac{t^{\frac{2}{\kappa}} - \beta_1}{\alpha_1} \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where the inequality is equivalent to

$$\frac{t^{\frac{2}{\kappa}} - \beta_1}{\alpha_1} \geq 0 \Rightarrow t^{\frac{2}{\kappa}} - \beta_1 \geq 0 \Rightarrow t^{\frac{2}{\kappa}} \geq \beta_1 \Rightarrow t \geq \beta_1^{\frac{\kappa}{2}}$$

□

The non-negativity of the process $Y = (\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}}$ is key in many ideas related to the tail index equation. In fact, it is advantageous to use this idea to simplify our expression for the expectation through the next lemma.

Lemma 2.12. *Given a random variable $Y \sim F_Y$ that is almost surely non-negative. The expectation can be defined by*

$$\mathbb{E}(Y) = \int_0^\infty (1 - F_Y(t)) dt$$

Proof. Let f_Y represent the PDF of Y . With this definition along with the definition of F_Y and the expectation, the proof can be seen directly as follows

$$\begin{aligned}
\int_0^\infty (1 - F_Y(t)) dt &= \int_0^\infty (1 - \mathbb{P}(Y \leq t)) dt \\
&= \int_0^\infty \mathbb{P}(Y > t) dt \\
&= \int_0^\infty \int_t^\infty f_Y(y) dy dt \\
&= \int_0^\infty \int_0^y dt f_Y(y) dy \\
&= \int_0^\infty y f_Y(y) dy \\
&= \mathbb{E}(Y)
\end{aligned}$$

□

Through Lemma 2.12, the direct evaluation of the equality can be produced with respect to α_1 and β_1 .

Theorem 2.13. *Let Z be a continuous random variable with symmetrical distribution F_Z . Then for $\alpha_1, \beta_1, \kappa > 0$ the expectation of $Y = (\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}}$ can be defined by*

$$\mathbb{E}(Y) = \beta_1^{\frac{\kappa}{2}} + 2 \int_{\beta_1^{\frac{\kappa}{2}}}^\infty F_Z \left(-\sqrt{\frac{t^{\frac{2}{\kappa}} - \beta_1}{\alpha_1}} \right) dt$$

Proof. In Theorem 2.11 it was shown that the distribution $\alpha_1 Z^2 + \beta_1$ was almost surely non-negative. As a consequence $\forall \alpha_1, \beta_1, \kappa > 0$ it can be shown that

$$\mathbb{P}(\alpha_1 Z^2 + \beta_1 \geq 0) = 1 \Rightarrow \mathbb{P}((\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}} \geq 0) = 1$$

and therefore $Y = (\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}}$ is also almost surely non-negative. Therefore, Lemma 2.12 can be applied to find the expectation

$$\begin{aligned}
\mathbb{E}(Y) &= \int_0^\infty 1 - F_Y(t) dt \\
&= \int_0^{\beta_1^{\frac{\kappa}{2}}} 1 - F_Y(t) dt + \int_{\beta_1^{\frac{\kappa}{2}}}^\infty 1 - F_Y(t) dt \\
&= \int_0^{\beta_1^{\frac{\kappa}{2}}} 1 dt + \int_{\beta_1^{\frac{\kappa}{2}}}^\infty 1 - 2F_Z \left(\sqrt{\frac{t^{\frac{2}{\kappa}} - \beta_1}{\alpha_1}} \right) + 1 dt \\
&= \beta_1^{\frac{\kappa}{2}} + 2 \int_{\beta_1^{\frac{\kappa}{2}}}^\infty 1 - F_Z \left(\sqrt{\frac{t^{\frac{2}{\kappa}} - \beta_1}{\alpha_1}} \right) dt \\
&= \beta_1^{\frac{\kappa}{2}} + 2 \int_{\beta_1^{\frac{\kappa}{2}}}^\infty F_Z \left(-\sqrt{\frac{t^{\frac{2}{\kappa}} - \beta_1}{\alpha_1}} \right) dt
\end{aligned}$$

where the distribution of Y was from Theorem 2.11 and the final step was due to the symmetry of Z . $\square$

Theorem 2.13 inspires a continuous function $f(x) : \mathbb{R}^+ \rightarrow \mathbb{R}$ with

$$f(x) = \mathbb{E}((\alpha_1 Z^2 + \beta_1)^{\frac{x}{2}}) - 1 = \beta_1^{\frac{x}{2}} + 2 \int_{\beta_1^{\frac{x}{2}}}^{\infty} F_Z \left(-\sqrt{\frac{t^{\frac{2}{x}} - \beta_1}{\alpha_1}} \right) dt - 1$$

that has root when $\mathbb{E}((\alpha_1 Z^2 + \beta_1)^{\frac{x}{2}}) = 1$, that is, when $x = \kappa$ is the tail index. Via root finding algorithms, this function makes it possible to determine the tail index of the GARCH(1,1) process accurately using only α_1, β_1 for the case of Gaussian innovations.

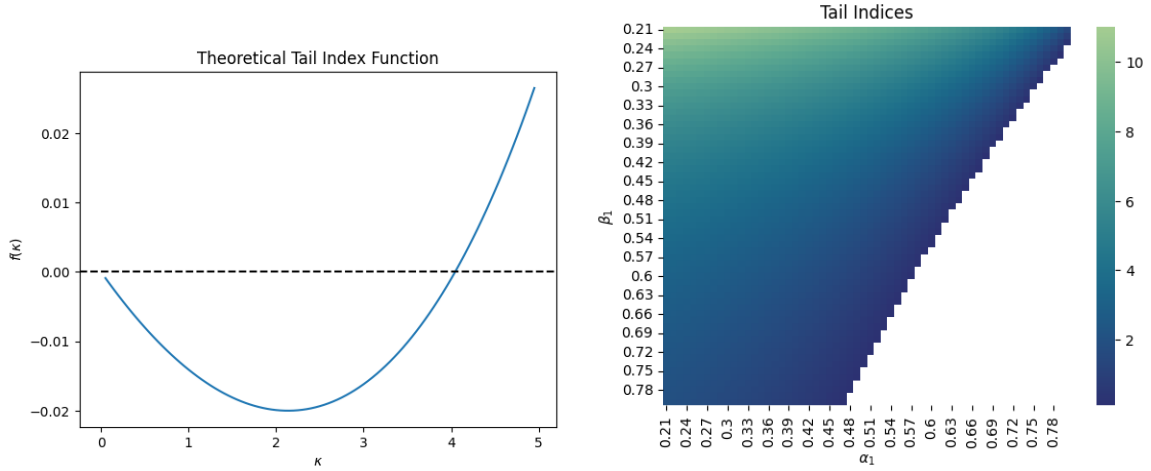


Figure 7: Left: Defined function $f(x)$ for $\alpha_1 = 0.1386, \beta_1 = 0.8415$ and Gaussian innovations. Right: Heatmap of tail index values for ranges of $\alpha_1, \beta_1 \in [0.2, 0.8]$

The heatmap in Figure 7 visualises the relationship with the two parameters α_1, β_1 and the tail index κ . As an approximation, the tail index κ decreases as $\alpha_1 + \beta_1$ increases. As $\alpha_1 + \beta_1 \rightarrow 1$ from below, we have $\kappa \rightarrow 2$ from above, showing that the condition $\alpha_1 + \beta_1 < 1$ enforces $\kappa < 2$ and therefore covariate stationarity. We also have $\kappa \rightarrow 0$ as we approach the stationarity barrier, indicating extremely heavy tails.

3 Distributional Forecasting using GARCH

This section implements the GARCH(1,1) model to capture time-varying volatility in financial log-returns and evaluating its efficacy. Since GARCH configurations only estimate conditional volatility, a separate process is required to estimate the mean of the distribution and produce a full probabilistic forecast. Multiple mean processes will be considered

- Constant Mean
- Autoregressive (AR(1)) Model [21]
- Moving Average (MA(1)) Model [22]
- ARMA(1,1) Model [23]

each with and without the GARCH component to model its volatility, allowing for a direct comparison of the contribution of the heteroskedasticity approximation. Each model pair will fit a one-step forecast via maximum likelihood on 1000 consecutive daily log-returns of Bitcoin (BTC-USD) up to and including 31st December 2025. The choice of Bitcoin was as a representative of the cryptocurrency market, as its most liquid and widely traded asset. This market is known to contain more extreme volatilities, so improving asymptotic tail estimations is important for risk-sensitive applications. A test set of 1000 data points was chosen to represent long-term trends in the returns including many market rises/falls that will ensure robustness in these situations.

Each forecast will be trained recursively using a rolling sample before each specific data point. The maximum likelihood GARCH parameters (α_1, β_1) will be re-estimated each iteration. This choice adds complexity to the algorithm so computational cost becomes a factor to consider. The size of this sample also introduces an interesting trade-off. A sample size too small would not be representative of the distribution and thus would produce non-meaningful forecasts. A sample size too large may include redundant data that is not representative of the distribution since taking log-returns only induces approximate stationarity. Therefore, we implement a Monte-Carlo simulation to investigate the causal bias on the parameter estimates for a range of sample sizes $n \in \{100, \dots, 1000\}$. An appropriate estimate for the GARCH parameters for Bitcoin daily log-returns is $\alpha_0 = 0.0001, \alpha_1 = 0.1035, \beta_1 = 0.8650$ [24] so we create a random sample with these parameters and calculate the mean squared error of the estimated parameters over 500 iterations.

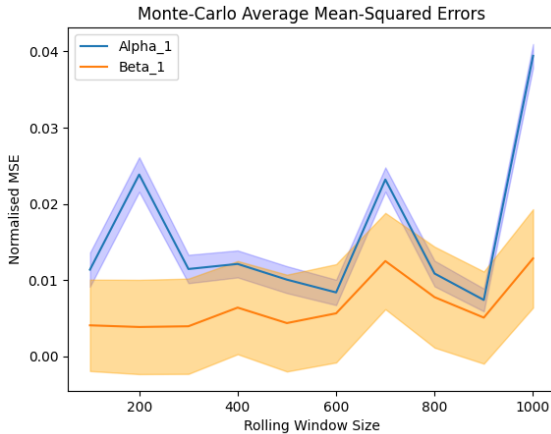


Figure 8: Normalised average mean squared errors for 1000 maximum likelihood estimates of GARCH(1,1) parameters $\alpha_1 = 0.1035, \beta_1 = 0.8650$ as a function of rolling window size n .

The results support the trade-off claim, since neither α_1 nor β_1 display a monotonic curve. The confidence intervals tell us that while α_1 does display a higher bias, its estimation is generally more stable than β_1 . The results dictate a high estimation error and instability in the region $n \in [100, 300]$. This reflects on insufficient data to capture the trends reliably. As the window size increases to $n \in [300, 600]$, we see small decreases in the errors of both parameters and an increase in stability. Notably, beyond $n = 500$ we observe no consistent improvement suggesting that the extra information is not representative of the distribution. This justifies a choice of $n = 500$ as a reasonable compromise for the rolling window size of our Bitcoin log-return forecasts.

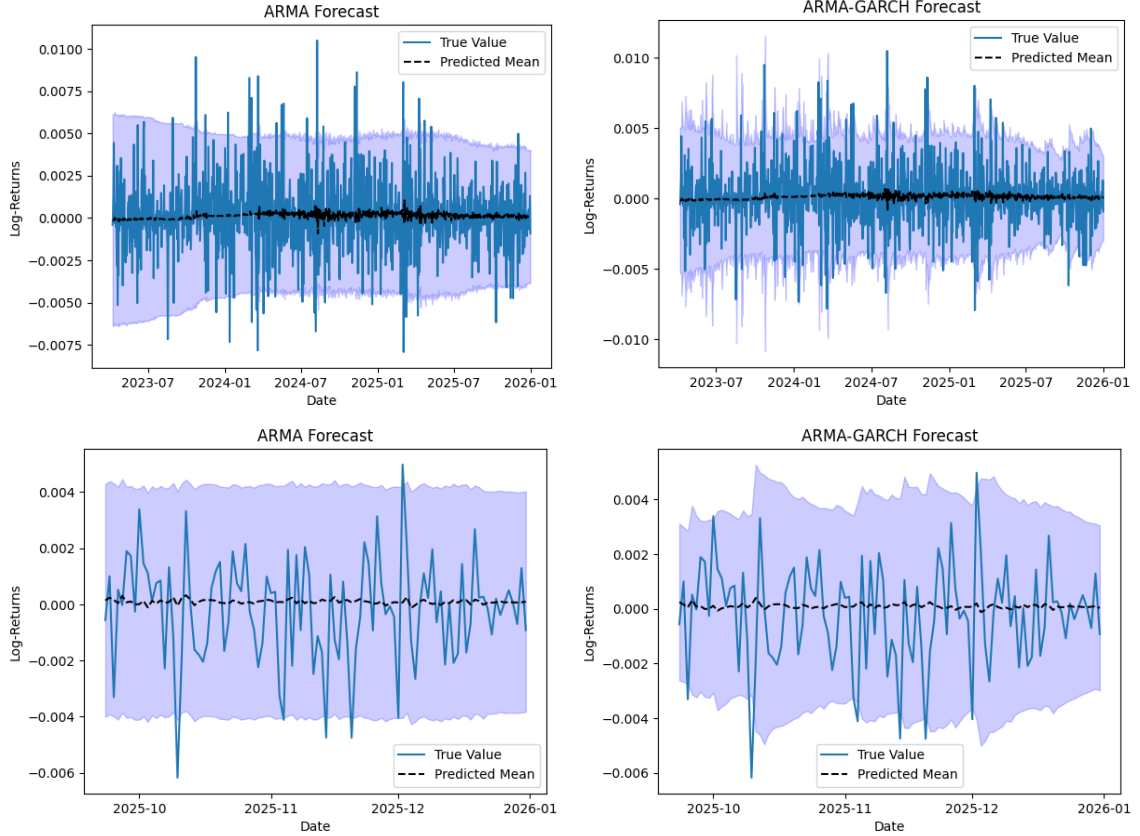


Figure 9: Rolling (500 window size) maximum likelihood forecasts for 1000 (Top) and 100 (Bottom) daily log-returns of BTC-USD using ARMA (Left) and ARMA-GARCH (Right) models up to and including 31st December 2025. The dotted line shows the predicted mean and the shaded region are 95% confidence intervals for the next day's log-returns.

The plots emphasise the effect of introducing GARCH innovations, which explicitly models conditional heteroskedasticity. The ARMA model assumes constant volatility so shows no evolution. On the other hand, ARMA-GARCH responds to large shocks in the returns visibly by creating wider bands. These remain elevated for some time and shrink gradually. This is exactly the persistence and simulated long-term memory which we have shown GARCH processes to exhibit. The visualisations highlight the true nature of the GARCH framework: the returns estimate itself is not

improved, rather its uncertainty over time.

Alternatively, we will estimate the effects of constraining the tail index of the process, in hopes that this will characterise distribution the tails better. This process will involve deriving empirical estimates of the tail index and constraining the parameters to the tail index is exhibited.

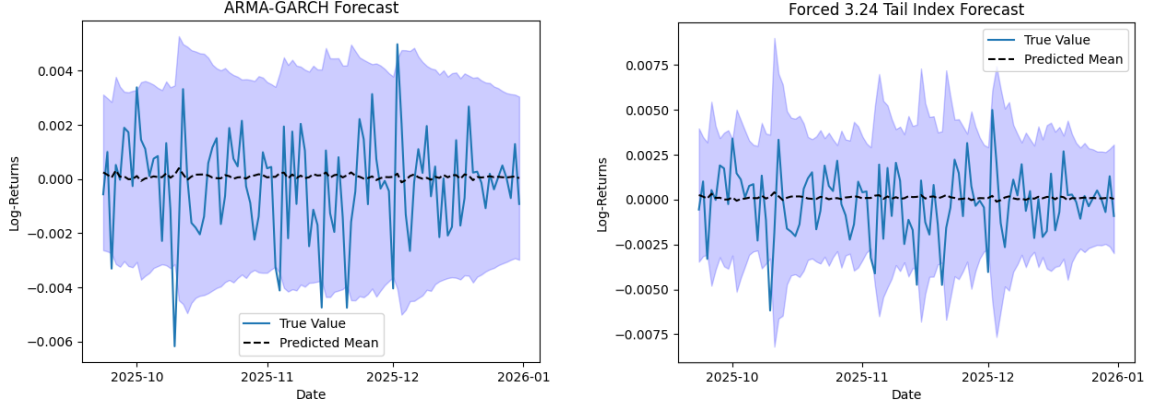


Figure 10: Rolling (500 window size) maximum likelihood forecasts for 100 daily log-returns of BTC-USD using ARMA-GARCH up to and including 31st December 2025. Optimisation is constrained to $\alpha_1, \beta_1 \in [0, 1]$ (Left) and α_1, β_1 such that $\kappa = 3.24$ (Right)

As discussed above, the ARMA-GARCH captures clusters in volatility with an increase and slow decay of variance after large shocks. In contrast, Figure 10 shows that constraining the choice of α_1, β_1 such that we have a tail index $\kappa = 3.24$ makes the spikes in uncertainty more pronounced, that is, we have a greater allowance for extreme returns. Consequently, this gives us two distinct methods for forecasting. Evaluations of these, including their weaknesses and contexts such that they thrive, will be explored in this section.

3.1 Model Evaluation

Each forecast created must be evaluated against one another to provide a comparative insight onto their predictive power. In practice, it is advantageous to consider proper scoring rules.

Definition 3.1 (Proper Scoring Rules [17]). *Let $\mathcal{Y}$ be a set of outcomes, $\mathcal{F}$ be a σ -algebra on $\mathcal{Y}$ and $\mathcal{P}$ denote a convex set of probability measures on the space $(\mathcal{Y}, \mathcal{F})$. A scoring rule is a function*

$$S : \mathcal{P} \times \mathcal{Y} \rightarrow \mathbb{R} \cup \{-\infty, \infty\}, \quad (\mathbb{P}, y) \mapsto S(\mathbb{P}, y)$$

such that $y \mapsto S(\mathbb{P}, y)$ is $\mathcal{P}$ -quasi-integrable for every $\mathbb{P} \in \mathcal{P}$. A scoring rule is called proper if

$$S(\mathbb{Q}, \mathbb{Q}) \leq S(\mathbb{P}, \mathbb{Q})$$

for every $\mathbb{P}, \mathbb{Q} \in \mathcal{P}$ and strictly proper if it holds with equality if and only if $\mathbb{P} = \mathbb{Q}$

The first known introduction of a (strictly) proper scoring rule was by Brier [25] as verification of probabilistic meteorological forecasts via the Brier score

$$S_{\text{Brier}}(\mathbb{P}, y) = (p - y)^2$$

where $p = \mathbb{P}(y = 1)$. The idea was to introduce a scoring rule which would discourage hedging by a dishonest forecaster. The concept was defined rigorously by McCarthy [26] and has since played a dominant role in belief elicitation within economics [27]. Without propriety, a forecaster may prioritise extreme cases to provoke confidence or reward specific values unfairly.

The first scoring metric that will be used for the purposes of this paper is the log-score

$$S_{\text{Log}}(\mathbb{P}, y) = -\log f_{\mathbb{P}}(y)$$

where $f_{\mathbb{P}}$ is the probability density of $\mathbb{P}$. Empirical risk minimisation for this scoring method is equivalent to maximum likelihood estimation. The log-score is known to be a local scoring rule; only the density at the outcome matters so extreme tail events are penalised massively. Thus, the log-score is useful to display the accuracy of a given forecast. It can be shown that the log-score is a strictly proper scoring function [28].

The second scoring rule is the continuous ranked probability score

$$S_{\text{CRPS}}(\mathbb{P}, y) = \int (F_{\mathbb{P}}(x) - \mathbb{1}\{y \leq x\})^2 dx$$

where $F_{\mathbb{P}}$ is the cumulative distribution of $\mathbb{P}$. It can also be seen as the Brier score integrated over $\mathbb{R}$ and has an equivalent representation

$$\begin{aligned} S_{\text{CRPS}}(\mathbb{P}, y) &= \int |x - y| d\mathbb{P}(x) - \frac{1}{2} \iint |x - x'| d\mathbb{P}(x) d\mathbb{P}(x') \\ &= \mathbb{E}|X - y| - \frac{1}{2} \mathbb{E}|X - X'| \end{aligned}$$

where X and X' are i.i.d. random variables with distribution $F_{\mathbb{P}}$. CRPS is also a strictly proper scoring rule [29] but is global, meaning the entire shape of the distribution is considered and penalised, rather than just the density at the outcome. This is a key difference between our two scores and will provide more insight into the reasoning behind a difference in score. For example, in the case of tail miscalibration the two scores will disagree.

Both scoring methods will be applied to each of the 1000 forecasts generated by each method. By averaging these scores over the interval, we can develop a general idea on the effectiveness of the predictive models. Note that a lower (more negative) score indicates a greater predictive performance for both scoring methodologies.

Mean Process	Without GARCH		With GARCH	
	Log-Score	CRPS	Log-Score	CRPS
Constant Mean	-4.656351	0.001208	-4.703490	0.001185
AR(1)	-4.653596	0.001211	-4.701922	0.001186
MA(1)	-4.654072	0.001210	-4.697571	0.001193
ARMA(1,1)	-4.652882	0.001212	-4.698435	0.001192

Table 1: Average Scores for Probabilistic Forecasts

Table 1 presents that when GARCH is used to model the volatility both the log-score and CRPS decrease across all mean specifications. When analysing log-scores, we see a negligible spread across each mean-model (≈ 0.003). This has the implication that the mean is not a dominant property of forecast performance. A similar conclusion can be made from the CRPS which also permits a negligible spread (≈ 0.00001).

However, a consistent difference can be seen when considering the addition of GARCH to each mean specification. In the case of the log-score each model witnesses a decrease by ≈ 0.05 . For the CRPS, there is also a consistent decrease ≈ 0.00002 . The significance of these differences will be evaluated in the next section. Now, we study the reason for this decrease by plotting the time series of the scores.

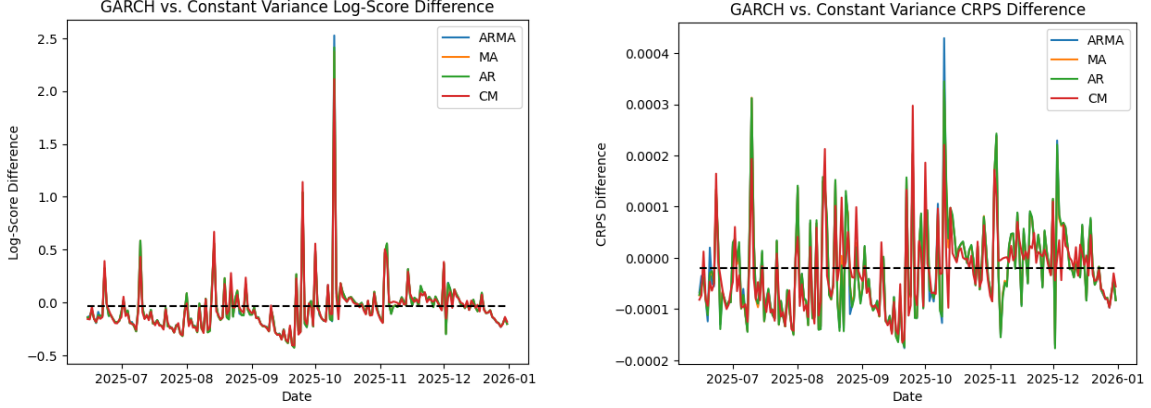


Figure 11: Differences in log-score (left) and CRPS (right) for the GARCH and constant volatility models over varied time. Positive values indicate that ARMA-GARCH produced a much better probabilistic forecast for that time point.

The results in Figure 11 show irregular but notable gains in both scores, with the CRPS being less dramatic. These are particularly prevalent in periods of high volatility, such as at the beginning of October 2025. The differences remain mixed and small in more stable regions. Therefore, the benefits of modelling conditional heteroskedasticity are most evident when its magnitude is large.

3.2 Significance Testing

It would be admirable to devise a test of the significance of the decrease in scores within each pair of processes. Suppose that $(\mathbb{P}_t)_{t \in \mathbb{Z}}$ and $(\mathbb{P}_t^*)_{t \in \mathbb{Z}}$ are two sequences of probabilistic forecasts, Diebold and Mariano [30] proposed a statistical test with the null hypotheses

$$H_0^- : \text{Proposed model } \mathbb{P} \text{ predicts at least as well as the standard model } \mathbb{P}^*$$

$$H_0^+ : \text{Proposed model } \mathbb{P} \text{ predicts at most as well as the standard model } \mathbb{P}^*$$

that can test for significant differences within the pairs of forecasts. Let $(x_t)_{t \in \mathbb{Z}}$ the true values of the log-returns, then the test statistic [31]

$$T = \frac{\Delta_n \bar{S}}{\hat{\sigma}_n / \sqrt{n}}, \quad \Delta_n \bar{S} = \frac{1}{n} \sum_{t=1}^n (S(\mathbb{P}_t, x_t) - S(\mathbb{P}_t^*, x_t))$$

can be used to produce an asymptotic η -level test of the hypotheses, where $\hat{\sigma}_n$ is a HAC-estimator [32] of the asymptotic variance $\hat{\sigma}_n = \text{Var}(\sqrt{n} \Delta_n \bar{S})$ and S is the scoring function. The η -level test is therefore

$$\Phi(T) \leq \eta \Rightarrow \text{Reject } H_0^+, \quad 1 - \Phi(T) \leq \eta \Rightarrow \text{Reject } H_0^-$$

which will provide a comparison between each pair of forecasts.

The Diebold-Mariano test relies on assumptions on the asymptotic behaviour and dependence in the structure of the scores. The use of proper scoring to construct the forecast evaluations for the Diebold-Mariano tests does not guarantee that these assumptions are satisfied to make a statistical inference. In particular, the loss process must exhibit stationarity and weak dependence. It does, however, provide us with comparisons that are well theoretically grounded and meaningful.

Mean Process	DM Test Statistic (T)	p-Value (H_0^+)	p-Value (H_0^-)
Constant Mean	-3.086648	0.001012	0.998988
AR(1)	-2.994505	0.001374	0.998626
MA(1)	-3.003437	0.001335	0.998665
ARMA(1,1)	-3.045694	0.001161	0.998839

Table 2: Diebold-Mariano Tests using Log-Score

The test produced consistently negative Diebold-Mariano statistics which provide enough evidence to reject the hypothesis H_0^+ at the 95% level. If the assumptions were satisfied, this would tell us that, according to the log-score, introducing GARCH to model the volatility makes the performance at least as good, if not better than without. Similar inferences can be made from the CRPS.

Mean Process	DM Test Statistic (T)	p-Value (H_0^+)	p-Value (H_0^-)
Constant Mean	-5.617545555	0.0000000096	0.9999999903
AR(1)	-4.6623197061	0.0000015633	0.9999984367
MA(1)	-4.8144118376	0.0000007382	0.9999992618
ARMA(1,1)	-4.6092924224	0.0000020202	0.9999979798

Table 3: Diebold-Mariano Tests using CRPS

In fact, the test statistics which use the CRPS are more negative than that of the log-score and so provide stronger evidence to reject H_0^+ . Though this method cannot be used to ensure a statistically significance, the strength of the results make it unlikely that the assumptions would overturn conclusions.

Surprisingly, it is also of note that the test statistic for the constant mean pair is much lower than the rest. One may infer that this difference simply tells us that the choice of mean process has much smaller effect when implementing GARCH for the volatility. In fact, we can roughly test this hypothesis by examining the test statistics between constant mean and ARMA(1,1) processes

$$\text{Without GARCH : } T = 1.44925$$

$$\text{With GARCH : } T = 0.90816$$

which show that the difference between the two is smaller when implementing GARCH to model the volatility.

3.3 Tail Index Estimation

It was proven in Section 2.3 that the tail index of a GARCH(1,1) process with Gaussian innovations depends only on the parameters α_1 and β_1 . Further, it can be shown that for a given tail index and β_1 there is a unique value of α_1 such that the process exhibits such tail index.

Theorem 3.2. *Let Z be a continuous random variable with symmetrical distribution F_Z . Then for given $\kappa > 2$ and $0 \leq \beta_1 \leq 1$ there exists a unique $\alpha_1 \in [0, 1]$ such that expectation of $X = (\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}}$ satisfies*

$$\mathbb{E}(X) = \mathbb{E}((\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}}) = 1$$

Proof. One can make use of the Intermediate Value Theorem with both boundary conditions. Firstly, when $\alpha_1 = 0$ then

$$\mathbb{E}(X) = \mathbb{E}(\beta_1^{\frac{\kappa}{2}}) = \beta_1^{\frac{\kappa}{2}} \leq 1$$

Conversely, for the boundary $\alpha_1 = 1$ it can be seen that

$$\mathbb{E}(X) = \mathbb{E}((Z^2 + \beta_1)^{\frac{\kappa}{2}})$$

If we consider the function $f : \mathbb{R} \mapsto \mathbb{R}$ with

$$\begin{aligned} f(z) &= (z + \beta_1)^{\frac{\kappa}{2}} \\ f'(z) &= \frac{\kappa}{2}(z + \beta_1)^{\frac{\kappa}{2}-1} \\ f''(z) &= \frac{\kappa}{2}\left(\frac{\kappa}{2} - 1\right)(z + \beta_1)^{\frac{\kappa}{2}-2} \end{aligned}$$

Since $\kappa > 2$ this function is strictly convex in Z^2 as $f''(Z^2)$ is non-negative. Therefore, we can apply Jensen's inequality

$$\mathbb{E}(X) = \mathbb{E}((Z^2 + \beta_1)^{\frac{\kappa}{2}}) \geq (1 + \beta_1)^{\frac{\kappa}{2}} \geq 1$$

and thus the Intermediate Value Theorem applies and there exists an $\alpha_1 \in [0, 1]$ such that the equality holds.

Finally, for uniqueness, we can differentiate with respect to α_1

$$\begin{aligned} g(\alpha_1) &= \mathbb{E}((\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}}) \\ g'(\alpha_1) &= \mathbb{E}\left(\frac{\kappa}{2} Z^2 (\alpha_1 Z^2 + \beta_1)^{\frac{\kappa}{2}-1}\right) > 0 \end{aligned}$$

to see that $\mathbb{E}((\alpha_1 Z_t^2 + \beta_1)^{\frac{\kappa}{2}})$ is a monotone increasing function in α_1 and therefore the solution found earlier is unique. $\square$

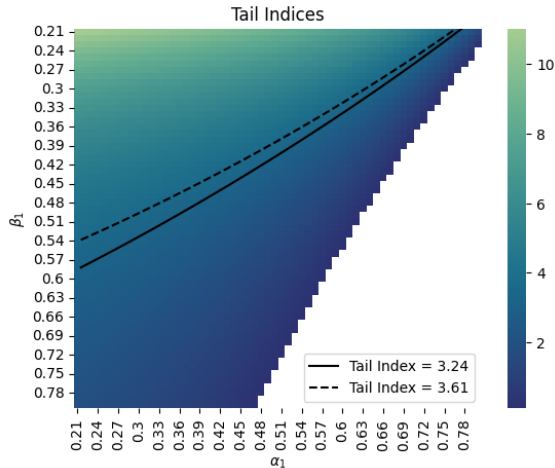


Figure 12: Heatmap representing the tail index of a GARCH(1,1) distribution for varied α_1 and β_1 with Gaussian innovations. The curve represents the points where the tail index κ is equal to 3.24 and 3.61

By limiting our maximum likelihood optimisation to this curve, the tail index of our forecasted distributions will be fixed. The evaluations of said distributions will provide interesting insights into the properties of the asymptotic tails over time.

Firstly, we introduce various methods to estimate the tail index of our sample. These estimators rely on the assumption of i.i.d observations with heavy tail asymptotic. Financial returns do exhibit dependence in the volatility which may affect the reliability of the approximation. For an ordered sample $X_1 \geq \dots \geq X_n$ it is possible to make use of Hill estimation [33] to produce an approximate value for $\gamma = \frac{1}{\kappa}$

$$H_{1,k} = \frac{1}{k} \sum_{i=1}^k \log \frac{X_i}{X_{k+1}}$$

for some $k \in \{1, \dots, n-1\}$.

The choice of k introduces a bias-variance trade-off. For small k , the estimator has a high variance due to the small sample size. For large k , the increase in observations from the central part of the model introduces bias by violating the asymptotic tail assumption. Hence, the accuracy of the estimation is highly sensitive to the choice of k . One determines this parameter is by graphing $\{(k, H_{1,k}^{-1}), 1 \leq k \leq n-1\}$, the so-called 'Hill-Plot' [34]. The area of the hill plot that is the most stable is said to be the best indication of the tail index. In practice, Hill plots are often highly unstable, especially for financial data that violates the i.i.d. assumption. This can make the stable region hard to determine and potentially unreliable, so one may make use of logarithmic scales (known as 'adjusted' Hill estimators) and smoothing [35] to better interpret the plot. This method is heuristic and may not produce a globally optimal estimate. Alternative approaches such as bootstrap methods [35] may produce a more robust selection.

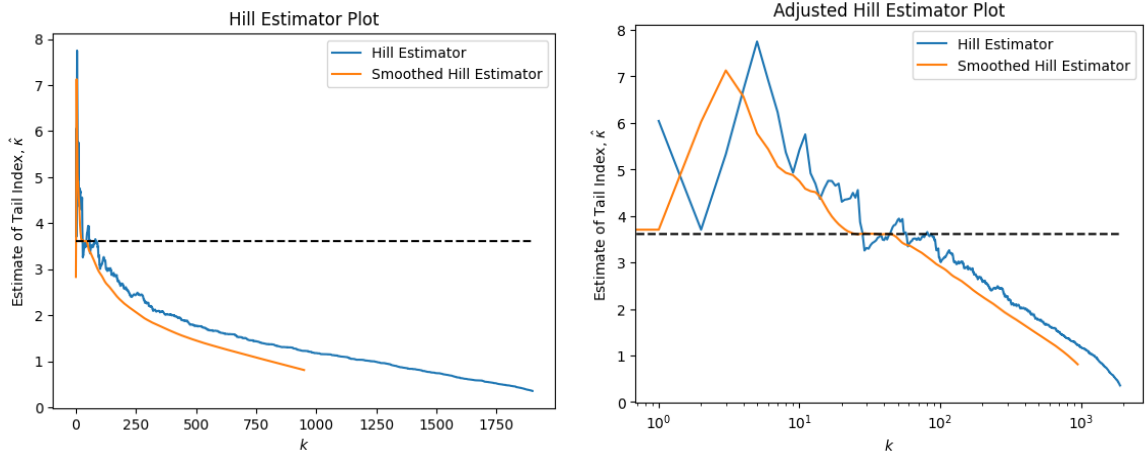


Figure 13: Hill plot (Left) and adjusted hill plot (Right) of the Hill Estimator for the log-returns of Bitcoin over the ten year interval between January 1st 2016 and January 1st 2026.

An estimate of $\hat{\kappa} = 3.61$ was produced by identifying the window of size 5 with the smallest variance ($k = 41$). This value is shown on the plots by the horizontal dashed line. The heuristic nature of this estimate, however, introduces uncertainty and potential unreliability.

Alternatively, by defining

$$H_{2,k} = \frac{1}{k} \sum_{i=1}^k \left(\log \frac{X_i}{\bar{X}_{k+1}} \right)^2$$

one can also make use of Moment estimation [36]

$$\hat{\gamma}_k = H_{1,k} + 1 - \frac{1}{2} \left(1 - \frac{H_{1,k}^2}{H_{2,k}} \right)^{-1}$$

which would provide us with an estimated tail index $\hat{\kappa} = \frac{1}{\hat{\gamma}}$ of the time series. Similarly, the choice of k is important in this calculation and must be chosen via the section of the Hill plot $\{(k, \gamma_k^{-1}), 1 \leq k \leq n-1\}$ with the most stability. We can once again make use of smoothing and logarithmic scales to better interpret these graphs.

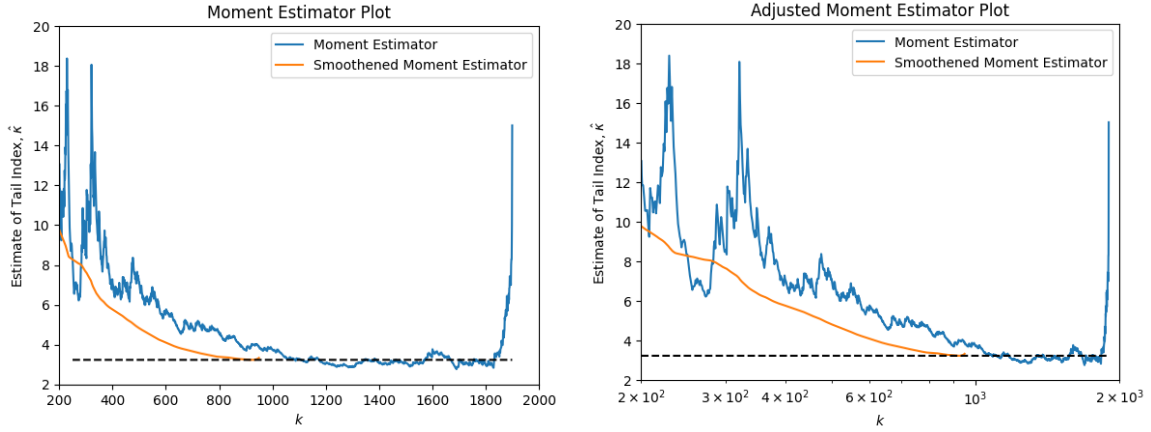


Figure 14: Hill plot (Left) and adjusted hill plot (Right) of the Moment Estimator for the log-returns of Bitcoin over the ten year interval between January 1st 2016 and January 1st 2026.

The window of size 5 with the smallest variance was found at $k = 925$, resulting in an estimated tail index of $\hat{\kappa} = 3.24$. The difference between this and the Hill estimate derives from sensitivity to the choice of threshold and the influence of higher order moments.

Both estimates of tail index value indicate substantial tail heaviness in the distribution, with the Moment estimator suggesting heavier than its counterpart. While we have finite variance and skewness, higher order moments are unstable and divergent. This is in line with the risk profiling of cryptocurrency markets as it implies a significant probability of extreme values.

Both approximations can be used as the choice of tail index to force onto the GARCH forecasts by maximising the likelihood function over its curve of α_1 and β_1 values. These forecasts will use ARMA(1,1) as a mean process and be compared to the regular ARMA(1,1) and ARMA-GARCH implementations from above.

Mean Process	Log-Score	CRPS
ARMA(1,1)	-4.652882	0.001212
ARMA-GARCH (Maximum Likelihood)	-4.698435	0.001192
ARMA-GARCH ($\kappa = 3.24$)	-4.647382	0.001170
ARMA-GARCH ($\kappa = 3.61$)	-4.640116	0.001169

Table 4: Average Scores for Probabilistic Forecasts

Interestingly, the increase in the average of the log-score indicates a worse performance whereas the decrease in CRPS advises a better performance. The log-score framework considers only the distributions at the observed values of the log-returns, meaning a higher value tell us that there is a lower probability density at these observed values only. Conversely, the CRPS penalises across the entire distribution and a lower score results from a better overall calibration of the model. This highlights a trade-off between local accuracy and global calibration. For example, risk estimation procedures such as VaR and expected shortfall rely on accurate tail modelling so the distributional benefits of the constrained tail index model may outweigh the log-score deterioration.

Between the two choices of tail indices we do not see a large change in scores. Thus, we conclude that the scores are relatively insensitive to choice of tail index, and that the choice of model is a much more important factor.

4 Discussion and Future Work

Overall, this research has produced empirical evidence that introducing the GARCH framework to model conditional volatility significantly improves predictive performance on the distributional forecasts of daily log-returns of Bitcoin. In particular, this change leads to more accurate estimates of forecast uncertainty, rather than of the mean estimate itself. Furthermore, imposing constraints on the parameters of the model to force a specific tail index introduces a trade-off between local accuracy and global calibration. Tail-constrained models are able to better capture extreme outcomes, which makes them relevant for practical applications where tail risk is of large concern. Though these results are explicit and can be explained heuristically, there may be some limitations present in this research.

- This analysis was restricted to modelling Bitcoin log-returns only. While this choice was made due to the authority of Bitcoin in the cryptocurrency market, one could question the generalisability of the results to other cryptocurrencies and markets. The volatility, for example, of cryptocurrency is much greater than that of traditional assets so may amplify the benefits we have witnessed.
- The tail index estimations that were produced were extremely sensitive to the choice of threshold k , meaning potential uncertainty would be present. Alternative estimators such as Pickands [37] or Peaks-Over-Threshold methods [38] could be thinkable. However, the results of this research did present an insensitivity in the distributional power of the forecasts to the choice of tail index κ .
- We have made the assumption of light-tailed Gaussian innovations which may underestimate extremes. Heavier-tailed alternative such as Student's-t could be of consideration. This would require significant changes to the tail index equation presented in Section 2.3.
- Specific augmentations to the GARCH configuration itself could also be possible extensions or generalisations. Examples could include investigating GARCH(p, q) with $p, q \neq 1$ or alternative models like E-GARCH [39]. This example in particular would allow asymmetric responses to positive and negative shocks which could be present in Bitcoin returns.

As suggestions for future work, one could consider optimising different statistics over the constraint curve. For example, optimising the CRPS may remove reliance on tail index estimation altogether and therefore the estimation error that comes with it. Empirical studies [40] also suggest that the tail index of Bitcoin is time-varying, so introducing rolling or adaptive estimation of the tail index may also be conceivable.

References

- [1] Fama, E.F., 1991. Efficient Capital Markets: II. The Journal of finance (New York) [Online], 46(5), pp.1575–1617. Available from: <https://doi.org/10.1111/j.1540-6261.1991.tb04636.x>.
- [2] Fama, E.F., 1965. The Behavior of Stock-Market Prices. The Journal of business (Chicago, Ill.) [Online], 38(1), pp.34–105. Available from: <https://doi.org/10.1086/294743>.
- [3] Black, F. and Scholes, M., 1973. The Pricing of Options and Corporate Liabilities. The Journal of political economy [Online], 81(3), pp.637–654. Available from: <https://doi.org/10.1086/260062>.
- [4] Campbell, J.Y., Lo, A.W. and MacKinlay, A.C., 1997. The econometrics of financial markets [Online]. Princeton, New Jersey: Princeton University Press. Available from: <https://ebookcentral.proquest.com/lib/bath/detail.action?docID=5515221>.
- [5] Aroussi, R., 2026. Yahoo! Finance market data downloader [Online]. Available from: <https://pypi.org/project/yfinance/> [Accessed 6 March 2026].
- [6] Cont, R., 2001. Empirical properties of asset returns: stylized facts and statistical issues. Quantitative finance [Online], 1(2), pp.223–236. Available from: <https://doi.org/10.1080/713665670>.
- [7] Buraczewski, D., Damek, E. and Mikosch, T., 2016. Stochastic Models with Power-Law Tails: The Equation $X = AX + B$ [Online]. Ed. by E. Damek and T. Mikosch. 1st ed. Cham: Springer Nature. Available from: <https://doi.org/10.1007/978-3-319-29679-1>.
- [8] Engle, R.F., 1982. Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation. Econometrica [Online], 50(4), pp.987–1007. Available from: <https://doi.org/10.2307/1912773>.
- [9] Bollerslev, T., 1986. Generalized autoregressive conditional heteroskedasticity. Journal of econometrics [Online], 31(3), pp.307–327. Available from: [https://doi.org/10.1016/0304-4076\(86\)90063-1](https://doi.org/10.1016/0304-4076(86)90063-1).
- [10] Jorion, P., 2008. Value at risk: the new benchmark for managing financial risk [Online]. 3rd ed. New York; McGraw-Hill. Available from: <https://ebookcentral.proquest.com/lib/bath/detail.action?docID=6262400>.
- [11] Duan, J.-C., 1995. THE GARCH OPTION PRICING MODEL. Mathematical finance [Online], 5(1), pp.13–32. Available from: <https://doi.org/10.1111/j.1467-9965.1995.tb00099.x>.

- [12] Box, G.E.P., Jenkins, G.M. and Reinsel, G.C., 2008. Time series analysis: forecasting and control. Fourth edition. Hoboken, N.J: Wiley, a John Wiley & Sons, Inc.
- [13] Cont, R., Potters, M. and Bouchaud, J.-P., 1997. Scaling in stock market data: stable laws and beyond [Online]. Available from: <https://doi.org/10.48550/arxiv.cond-mat/9705087>.
- [14] Mandelbrot, B., 1963. The Variation of Certain Speculative Prices. The Journal of business (Chicago, Ill.) [Online], 36(4), pp.394–419. Available from: <https://doi.org/10.1086/294632>.
- [15] Resnick, S.I., 2007. Heavy-Tail Phenomena: Probabilistic and Statistical Modeling [Online]. 1st ed. 2007. New York, NY: Springer New York. Available from: <https://doi.org/10.1007/978-0-387-45024-7>.
- [16] Kesten, H., 1973. Random difference equations and Renewal theory for products of random matrices. Acta mathematica [Online], 131, pp.207–248. Available from: <https://doi.org/10.1007/BF02392040>.
- [17] Waghmare, K. and Ziegel, J., 2025. Proper Scoring Rules for Estimation and Forecast Evaluation. Annual review of statistics and its application [Online]. Available from: <https://doi.org/10.1146/annurev-statistics-042424-050626>.
- [18] Goldie, C.M. and Maller, R.A., 2000. Stability of Perpetuities. The Annals of probability [Online], 28(3), pp.1195–1218. Available from: <https://doi.org/10.1214/aop/1019160331>.
- [19] Goldie, C.M., 1991. Implicit Renewal Theory and Tails of Solutions of Random Equations. The Annals of applied probability [Online], 1(1), pp.126–166. Available from: <https://doi.org/10.1214/aoap/1177005985>.
- [20] Breiman, L., 1965. On Some Limit Theorems Similar to the Arc-Sin Law. Theory of probability and its applications [Online], 10(2), pp.323–331. Available from: <https://doi.org/10.1137/1110037>.
- [21] Yule, G.U., 1926. Why do we Sometimes get Nonsense-Correlations between Time-Series?—A Study in Sampling and the Nature of Time-Series. Journal of the Royal Statistical Society [Online], 89(1), pp.1–63. Available from: <https://doi.org/10.2307/2341482>.
- [22] Slutsky, E., 1937. The Summation of Random Causes as the Source of Cyclic Processes. Econometrica [Online], 5(2), pp.105–146. Available from: <https://doi.org/10.2307/1907241>.
- [23] Box, G.E.P., Jenkins, G.M. and Reinsel, G.C., 2008. Time series analysis: forecasting and control. Fourth edition. Hoboken, N.J: Wiley, a John Wiley & Sons, Inc.

- [24] Venter, P.J. and Maré, E., 2020. GARCH generated volatility indices of Bitcoin and CRIX. *Journal of risk and financial management* [Online], 13(6), pp.1–15. Available from: <https://doi.org/10.3390/jrfm13060121>.
- [25] Brier, G.W., 1950. Verification of Forecasts Expressed in Terms of Probability. *Monthly weather review* [Online], 78(1), pp.1–3. Available from: [https://doi.org/10.1175/1520-0493\(1950\)078<0001:VOFEIT;2.0.CO;2](https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT;2.0.CO;2).
- [26] McCarthy, J., 1956. Measures of the Value of Information. *Proceedings of the National Academy of Sciences - PNAS* [Online], 42(9), pp.654–655. Available from: <https://doi.org/10.1073/pnas.42.9.654>.
- [27] Schotter, A. and Trevino, I., 2014. Belief Elicitation in the Laboratory. *Annual review of economics* [Online], 6(1), pp.103–128. Available from: <https://doi.org/10.1146/annurev-economics-080213-040927>.
- [28] Good, I.J., 1952. Rational Decisions. *Journal of the Royal Statistical Society. Series B, Methodological* [Online], 14(1), pp.107–114. Available from: <https://doi.org/10.1111/j.2517-6161.1952.tb00104.x>.
- [29] Matheson, J.E. and Winkler, R.L., 1976. Scoring Rules for Continuous Probability Distributions. *Management science* [Online], 22(10), pp.1087–1096. Available from: <https://doi.org/10.1287/mnsc.22.10.1087>.
- [30] Diebold, F.X. and Mariano, R.S., 2002. Comparing Predictive Accuracy. *Journal of business & economic statistics* [Online], 20(1), pp.134–144. Available from: <https://doi.org/10.1198/073500102753410444>.
- [31] Nolde, N. and Ziegel, J.F., 2017. Elicitability and Backtesting: Perspectives for Banking Regulation. *The annals of applied statistics* [Online], 11(4), pp.1833–1874. Available from: <https://doi.org/10.1214/17-aos1041>.
- [32] Andrews, D.W.K., 1991. Heteroskedasticity and Autocorrelation Consistent Covariance Matrix Estimation. *Econometrica* [Online], 59(3), pp.817–858. Available from: <https://doi.org/10.2307/2938229>.
- [33] Hill, B.M., 1975. A Simple General Approach to Inference About the Tail of a Distribution. *The Annals of statistics* [Online], 3(5), pp.1163–1174. Available from: <https://doi.org/10.1214/aos/1176343247>.

- [34] Drees, H., de Haan, L. and Resnick, S., 2000. How to Make a Hill Plot. The Annals of statistics [Online], 28(1), pp.254–274. Available from: <https://doi.org/10.1214/aos/1016120372>.
- [35] Resnick, S. and Stărică, C., 1997. Smoothing the Hill Estimator. Advances in applied probability [Online], 29(1), pp.271–293. Available from: <https://doi.org/10.1017/S0001867800027889>.
- [36] Dekkers, A.L.M., Einmahl, J.H.J. and De Haan, L., 1989. A Moment Estimator for the Index of an Extreme-Value Distribution. The Annals of statistics [Online], 17(4), pp.1833–1855. Available from: <https://doi.org/10.1214/aos/1176347397>.
- [37] Pickands, J., 1975. Statistical Inference Using Extreme Order Statistics. The Annals of statistics [Online], 3(1), pp.119–131. Available from: <https://doi.org/10.1214/aos/1176343003>.
- [38] Coles, S., 2001. An Introduction to Statistical Modeling of Extreme Values [Online]. 1st ed. 2001. London: Springer London. Available from: <https://doi.org/10.1007/978-1-4471-3675-0>.
- [39] Nelson, D.B., 1991. Conditional Heteroskedasticity in Asset Returns: A New Approach. Econometrica [Online], 59(2), pp.347–370. Available from: <https://doi.org/10.2307/2938260>.
- [40] Katsiampa, P., 2017. Volatility estimation for Bitcoin: A comparison of GARCH models. Economics letters [Online], 158, pp.3–6. Available from: <https://doi.org/10.1016/j.econlet.2017.06.023>.